

VINAYKUMAR B PATIL

EMBEDDED SYSTEMS ENGINEER

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SUMMARY

Embedded Systems Engineer with 2+ years of experience developing application and firmware software for aerospace, UAV (fixed-wing & agricultural drones), and IoT platforms. Proficient in C/C++, STM32/Cube Orange+ HAL/LL, FreeRTOS, and distributed embedded architectures. Skilled in multi-protocol communication (UART, SPI, I2C, CAN, Ethernet, Serial), flight control integration (PX4, ArduPilot), and open-source development. Experienced in cross-functional collaboration, defect lifecycle management, and delivering production-ready embedded systems.

TECHNICAL SKILLS

Languages: C, Embedded C, C++ (primary); C# (exposure)

Operating Systems: Embedded Linux (Ubuntu, OpenEmbedded, RHEL); FreeRTOS; QNX RTOS (familiar); VxWorks (familiar)

Flight Controls: PX4, ArduPilot (open-source); MAVLink C2 integration; fixed-wing & agricultural UAV platforms

Processors: ARM Cortex-M (STM32 HAL/LL, Cube Orange+), ESP32; PowerPC (familiar)

Protocols: UART, SPI, I2C, CAN, Serial, Ethernet, SBUS, PPM, MAVLink

Embedded Software: Distributed architecture, hardware abstraction layers, multi-protocol driver development, automated unit & regression testing, HIL validation

AI/ML: Exposure to AI/ML concepts for embedded applications (coursework)

Hardware: PCB Design (Altium, KiCad), board bring-up, logic analyzer, CRO, multimeter

Tools: STM32CubeIDE, Git, Jira, open-source libraries, peer code review

EXPERIENCE

Flying Wedge Defence & Aerospace

Feb 2024 – Present

Embedded Systems Engineer

- Developed embedded application firmware in C/C++ on STM32 & Cube Orange+ (ARM Cortex-M) for avionics, agricultural, and fixed-wing UAV platforms using FreeRTOS preemptive multitasking.
- Integrated PX4 and ArduPilot open-source flight control stacks; implemented MAVLink-based C2 communication and interfaced GPS, IMU, and RF peripherals.
- Built and maintained distributed embedded software architecture with modular HALs, multi-protocol drivers (UART, SPI, I2C, CAN, Ethernet, Serial), and inter-subsystem communication.
- Designed automated unit and regression test suites for C/C++ modules; validated firmware via hardware-in-loop testing, reducing integration defects through structured peer code reviews.
- Designed and validated sensing and power-monitoring PCBs (Altium/KiCad); performed board bring-up and hardware debugging for production prototypes.

Key Projects:

- Seed Dispenser System (Cube Orange+) — Developed precision dispensing firmware for agricultural UAV; implemented timed actuator control and CAN-based communication with flight controller.
- Grenade Launcher System (Cube Orange+) — Engineered payload release firmware with safety interlock logic and MAVLink command integration for defence UAV application.
- 2.4 GHz Wireless Transceiver for UAV Control — Designed transmitter-receiver system supporting 10 PWM channels with SBUS/PPM output; implemented SPI and I2C drivers for RF modules.
- Aeronero Air-to-Water System (Cube Orange+ / ESP32) — Integrated multi-sensor acquisition, real-time calibration, and Wi-Fi/Ethernet telemetry for a cloud-connected atmospheric water generation platform.
- Precision Current & Voltage Monitoring Module — High-accuracy measurement PCB with digital calibration firmware; validated against traceable references.

Bosch Global Software Technologies

Jan 2023 – May 2023

Software Development Engineer Intern

- Developed embedded application software for camera-based automotive systems on Linux; implemented real-time actuator control logic improving response latency and system reliability.
- Participated in defect tracking and resolution using SCM tools; performed peer code reviews following automotive coding standards and collaborated with cross-functional stakeholders.

EDUCATION

B.E. Electronics and Communication Engineering

2019 – 2023

KLE Technological University, Hubballi